Which total do we trust?

Name: _____ Write each answer **before** the slide reveals it. Changing an answer after the reveal is fine; cross it out, don't erase.

A. What does one row mean?

MIDTERM Q16

12 loans by 5 members; exactly 4 loans have `days_kept > 14`.

```
SELECT loan_id, branch, days_kept
FROM loans
WHERE days_kept > 14
ORDER BY days_kept DESC;
```

A1 Rows returned: _____ One row is one _____

B. Where does the key check go?

MIDTERM Q23

Each `students` row should be one student, but S07 may appear twice. The joined result has one row per tutoring session matched to a student.

```
SELECT COUNT(*), COUNT(DISTINCT student_id)
FROM _____ ;
```

B1 Circle one: **students** **joined result**

B2 Why not the other one?

C. Same rows, stored two ways

ROW VS COLUMN

The first six rows of the Pop-Tart export (all 7 fields). The query is `SELECT SUM(units) FROM sales;`

date	store_id	sku	product_name	flavor	units	unit_price
2026-08-22	S01	PT-STRW	Pop-Tarts Strawberry 8ct	Strawberry	17	3.49
2026-08-22	S01	PT-BSCN	Pop-Tarts Brown Sugar Cinnamon 8ct	Brown Sugar Cinnamon	9	3.29
2026-08-22	S01	PT-BLUE	Pop-Tarts Blueberry 8ct	Blueberry	7	3.49
2026-08-22	S02	PT-STRW	Pop-Tarts Strawberry 8ct	Strawberry	12	3.49
2026-08-22	S02	PT-BSCN	Pop-Tarts Brown Sugar Cinnamon 8ct	Brown Sugar Cinnamon	10	3.29
2026-08-22	S02	PT-BLUE	Pop-Tarts Blueberry 8ct	Blueberry	6	3.49

C1 Shade every cell the sum actually needs. Sum: _____

C2 Stored **row by row**, a reader passes through whole rows. Cells passed: _____ Stored **by column**, it can read just the units chunk. Cells read: _____

C3 How many different flavors are in the flavor column? _____ Give each one a number (0, 1, 2...) and rewrite the flavor chunk as numbers: _____

D. A correction arrives

SAFE UPDATES

One sales table is stored as files. A store resent one slice, so **C_new replaces C**; it is not extra sales. All four files are now in the folder.

file	units
A	100
B	200
C (old slice)	300
C_new (replacement)	280

D1 A query adds every file in the folder. Total: _____ Why is it wrong?

D2 The correct total after the correction: _____ Files counted: _____

D3 A report started before the correction was committed. What total should it finish with? _____

D4 If time: the writer crashed after writing C_new but before committing. What does a new reader get, and does C_new count? _____

E. Exit ticket: the campus shop

HAND THIS IN

A campus shop keeps its raw register logs **and** runs a daily sales report while corrections keep arriving. Same files as D, except this time C_new = **260**.

E1 Circle one and give one reason: warehouse | lake files only | lakehouse table | lake + warehouse

E2 What decides which files the report counts? _____

E3 New total: _____ Files counted: _____ Is file C still in storage? _____